LAB 3B

Advanced Cybersecurity Risk Quantification

Complete Solution | SecureBank Financial, Mobile Banking Platform

Field	Detail
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Date	31st October, 2026

> **Bottom line for the CISO and CFO**
> Current exposure is **$35.00M per year**. An **$805,000** investment in three controls removes **$26.96M (77.0%)** of that risk, a **33.5 : 1** benefit-to-cost ratio with payback in under a month. Approve all three controls, starting with the WAF immediately.

Total current exposure	Total Year 1 investment	Residual risk after controls
$35,000,000 / year	**$805,000**	**$8,037,500** / year

Phase 1: Risk Exposure Calculation

Task 1: Calculate SLE, ARO, and ALE

Formulas used

- **SLE** (Single Loss Expectancy) = Asset Value × Exposure Factor, or the given maximum or expected loss per incident.
- **ARO** (Annual Rate of Occurrence) = exploit probability, treated as the annual likelihood.
- **ALE** (Annualized Loss Expectancy) = SLE × ARO.

Finding 1: API Authentication Bypass

Component	Calculation	Value
SLE	Maximum single-incident loss (given)	$5,000,000
ARO	Exploit probability (given)	0.15
ALE	**$5,000,000 × 0.15**	**$750,000**

> Detection takes 48 hours, so one exploit could allow several unauthorized transfers before containment. The $5M maximum single-incident loss already reflects that exposure window.

Finding 2: Database Injection Vulnerability

Component	Calculation	Value
SLE	500,000 records × $250 per record	$125,000,000
ARO	Exploit probability (given)	0.25
ALE	**$125,000,000 × 0.25**	**$31,250,000**

> This is the dominant risk by a wide margin. One successful injection exposes the entire customer base.

Finding 3: Session Hijacking

Component	Calculation	Value
SLE	5,000 accounts × $1,500 per account	$7,500,000
ARO	Exploit probability (given)	0.40
ALE	**$7,500,000 × 0.40**	**$3,000,000**

Task 2: Risk Prioritization Matrix

Rank	Vulnerability	SLE	ARO	ALE	Risk priority
1	Database Injection	$125,000,000	0.25	$31,250,000	CRITICAL
2	Session Hijacking	$7,500,000	0.40	$3,000,000	HIGH
3	API Authentication Bypass	$5,000,000	0.15	$750,000	MEDIUM
	Total aggregate ALE			**$35,000,000 / yr**	

Prioritization thresholds

Priority	ALE range	Meaning
Critical	Greater than $10M	Catastrophic single-loss potential (regulatory, reputational)
High	$1M to $10M	Treat in the short term
Medium	$100K to $1M	Treat in the planned roadmap
Low	Below $100K	Accept or monitor

Task 3: Risk Visualizations

(a) Bar chart: ALE comparison

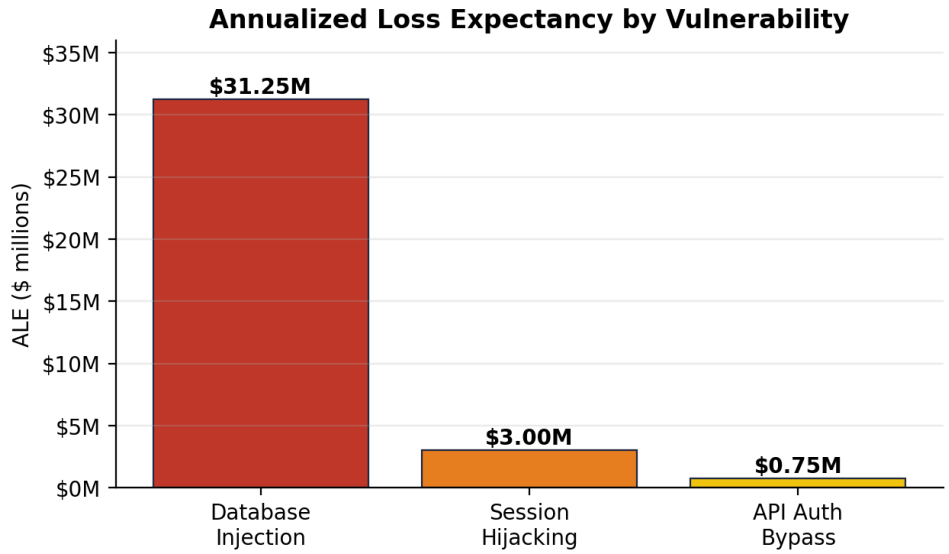


Figure 1. ALE by vulnerability, colored by priority (red, orange, yellow)

(b) Pie chart: risk distribution

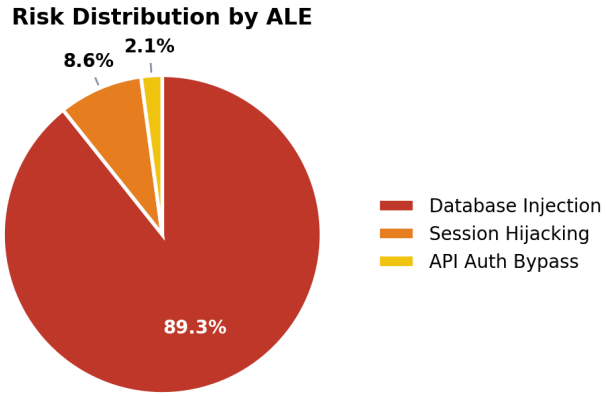


Figure 2. Share of total ALE by vulnerability

Vulnerability	ALE	% of total
Database Injection	$31.25M	89.3%
Session Hijacking	$3.00M	8.6%
API Authentication Bypass	$0.75M	2.1%
Total	**$35.00M**	**100.0%**

(c) Risk heat map: probability × impact

Probability ↓ Impact →	Low (below $1M)	Medium ($1M to <$10M)	High ($10M to $50M)	Critical (above $50M)
Very High (above 50%)				
High (25% to 50%)		**Session Hijacking** 40% | $7.5M		**Database Injection** 25% | $125M
Medium (10% to <25%)		**API Auth Bypass** 15% | $5M		
Low (below 10%)				

Color Codes: green = low, yellow = moderate, orange = high, red = critical.

> Database Injection sits in the **High probability / Critical impact** cell, so it is the top treatment priority. Session Hijacking (High probability, Medium impact) is second. API Authentication Bypass (Medium probability, Medium impact) is third.

Phase 2: Control Evaluation

Task 4: Cost-Benefit Analysis

Methodology

- **Year 1 total cost** = initial investment + annual maintenance.
- **Risk reduction ($)** = current ALE × effectiveness %.
- **Net Year 1 benefit** = risk reduction − Year 1 total cost.
- **ROI (Year 1)** = net Year 1 benefit ÷ Year 1 total cost × 100%.
- **Payback period** = initial investment ÷ (annual risk reduction − annual maintenance).

Control 1: Advanced API Security Gateway

Metric	Calculation	Value
Initial investment	Given	$350,000
Annual maintenance	Given	$50,000
Year 1 total cost	$350,000 + $50,000	$400,000
Risk reduction	90% × $750,000 (API ALE)	$675,000
Net Year 1 benefit	$675,000 - $400,000	$275,000
ROI (Year 1)	**$275,000 ÷ $400,000**	**68.75%**
Payback period	**$350,000 ÷ ($675,000 - $50,000)**	**0.56 years (about 6.7 months)**

> The conservative interpretation applies the 90% reduction to the API finding only. If the gateway also reduced Session Hijacking (which involves API and session tokens), the benefit would be larger.

Control 2: Web Application Firewall (WAF)

Metric	Calculation	Value
Initial investment	Given	$150,000
Annual maintenance	Given	$25,000
Year 1 total cost	$150,000 + $25,000	$175,000
Risk reduction	75% × $31,250,000 (Database Injection ALE)	$23,437,500
Net Year 1 benefit	$23,437,500 − $175,000	$23,262,500
ROI (Year 1)	**$23,262,500 ÷ $175,000**	**13,292.9%**
Payback period	**$150,000 ÷ ($23,437,500 − $25,000)**	**0.0064 years (about 2.3 days)**

> The WAF is extraordinarily cost-effective because it addresses the largest ALE. It is the clear must-buy.

Control 3: Multi-Factor Authentication Enhancement

Metric	Calculation	Value
Initial investment	Given	$200,000
Annual maintenance	Given	$30,000
Year 1 total cost	$200,000 + $30,000	$230,000
Risk reduction	95% × $3,000,000 (Session Hijacking ALE)	$2,850,000
Net Year 1 benefit	$2,850,000 − $230,000	$2,620,000
ROI (Year 1)	**$2,620,000 ÷ $230,000**	**1,139.1%**
Payback period	**$200,000 ÷ ($2,850,000 − $30,000)**	**0.071 years (about 26 days)**

Summary: control cost-benefit

Control	Initial	Maint.	Yr 1 cost	Risk reduction	Net benefit	ROI (Yr 1)	Payback
API Security Gateway	$350,000	$50,000	$400,000	$675,000	$275,000	68.75%	6.7 mo
WAF	$150,000	$25,000	$175,000	$23,437,500	$23,262,500	13,292.9%	2.3 days
MFA Enhancement	$200,000	$30,000	$230,000	$2,850,000	$2,620,000	1,139.1%	26 days
All three combined	**$700,000**	**$105,000**	**$805,000**	**$26,962,500**	**$26,157,500**	**3,249.4%**	**-**

> **Combined (all three controls):** Year 1 cost $805,000; total risk reduction $675,000 + $23,437,500 + $2,850,000 = $26,962,500; net benefit $26,157,500; aggregate ROI 3,249.4%.

Task 5: Control Selection Analysis

(a) Scatter plot: cost vs. effectiveness

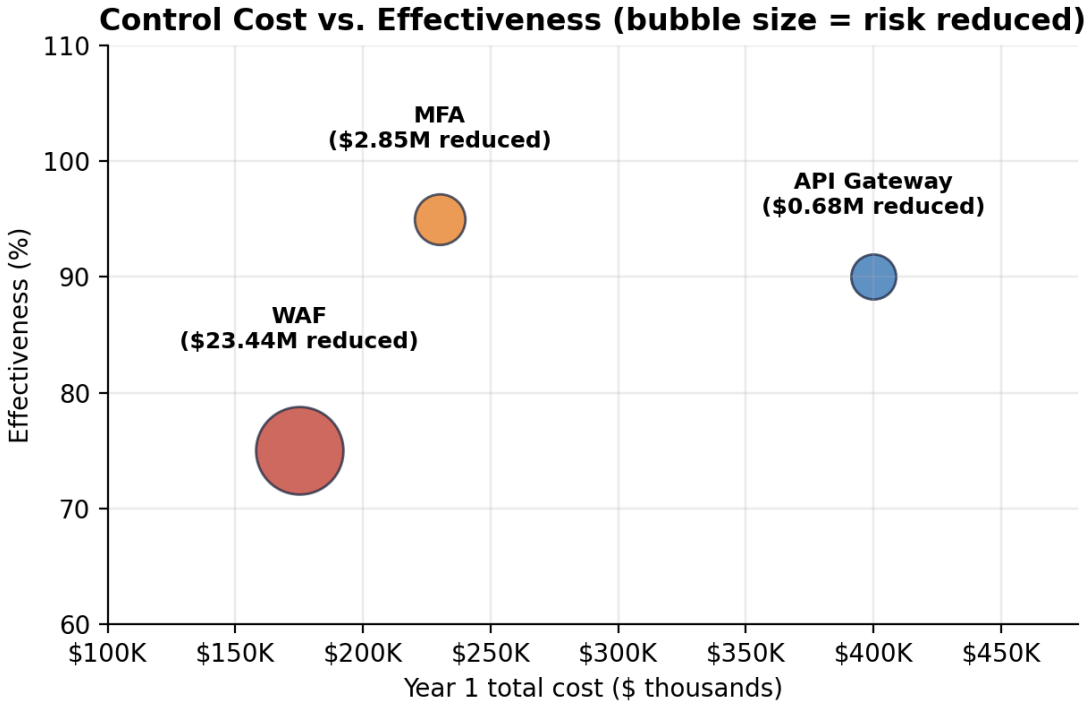


Figure 3. WAF and MFA offer the best value; the API Gateway is effective but costliest

(b) Bar chart: ROI comparison

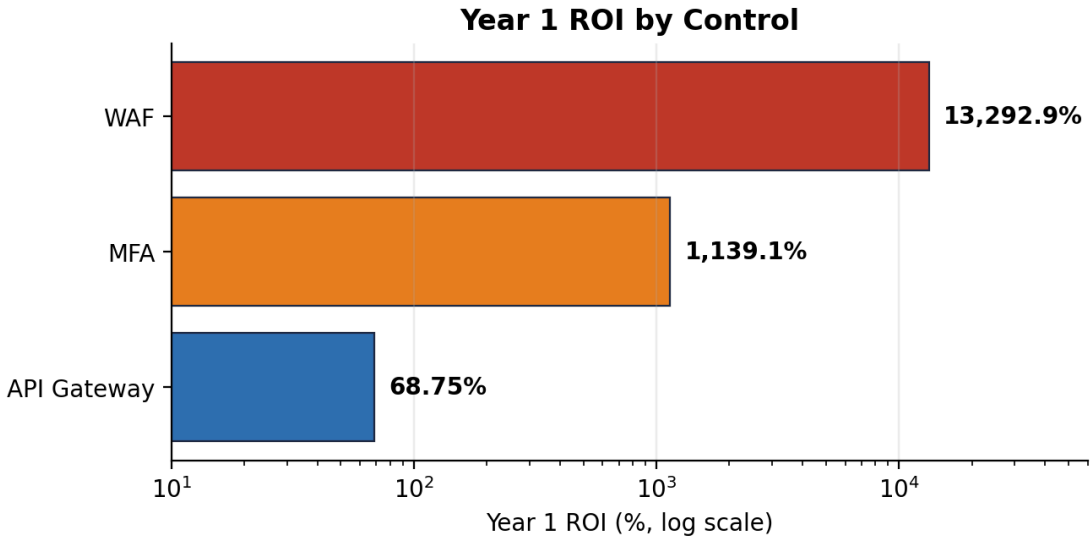


Figure 4. Year 1 ROI by control (log scale, because WAF ROI dwarfs the others)

(c) Line graph: risk reduction over time

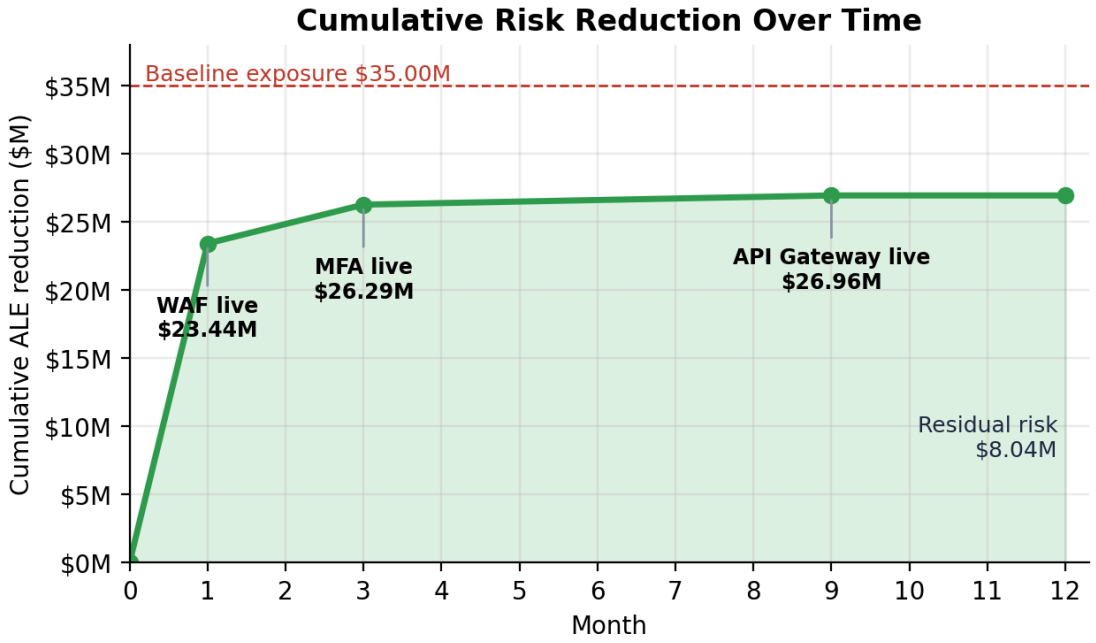


Figure 5. Cumulative ALE reduction as each control goes live

Month	Action	Cumulative ALE reduction
0	Baseline	$0
1	WAF deployed (immediate)	$23.44M
3	MFA deployed (90 days)	$26.29M
9	API Gateway fully deployed	$26.96M
12	Steady state	$26.96M

Phase 3: Executive Reporting

Task 6: Executive Dashboard (One-Page Summary)

SECUREBANK FINANCIAL: MOBILE BANKING RISK DASHBOARD

Prepared for: CISO and CFO | Classification: Confidential

Top 3 risks (current annualized loss expectancy)

#	Risk	ALE	Share of total	Priority
1	Database Injection	$31.25M	89.3%	CRITICAL
2	Session Hijacking	$3.00M	8.6%	HIGH
3	API Authentication Bypass	$0.75M	2.1%	MEDIUM
	Total current risk exposure	**$35.00M / year**	**100.0%**	

Recommended controls and investment

Control	Yr 1 cost	Risk reduced	ROI	Payback
WAF	$175,000	$23,437,500	13,293%	~2 days
MFA Enhancement	$230,000	$2,850,000	1,139%	~26 days
API Security Gateway	$400,000	$675,000	69%	~7 months
Total investment	**$805,000**	**$26,962,500**	**3,249%**	**< 1 month**

Expected risk reduction

Before controls	After controls (residual)	Risk eliminated
$35.00M / year	**$8.04M / year**	**$26.96M / year** (77.0% reduction)

> **Recommendation:** Deploy all three controls in Year 1. The $805K investment returns $26.96M in avoided losses, a 33.5 : 1 benefit-to-cost ratio. The WAF is the highest-urgency purchase because it protects 89% of total risk.

Residual risk check: Database Injection $31.25M × 25% = $7.81M; Session Hijacking $3.00M × 5% = $0.15M; API Bypass $0.75M × 10% = $0.075M. Total residual ALE = **$8.04M**, which equals $35.00M − $26.96M.

Required dashboard visualizations

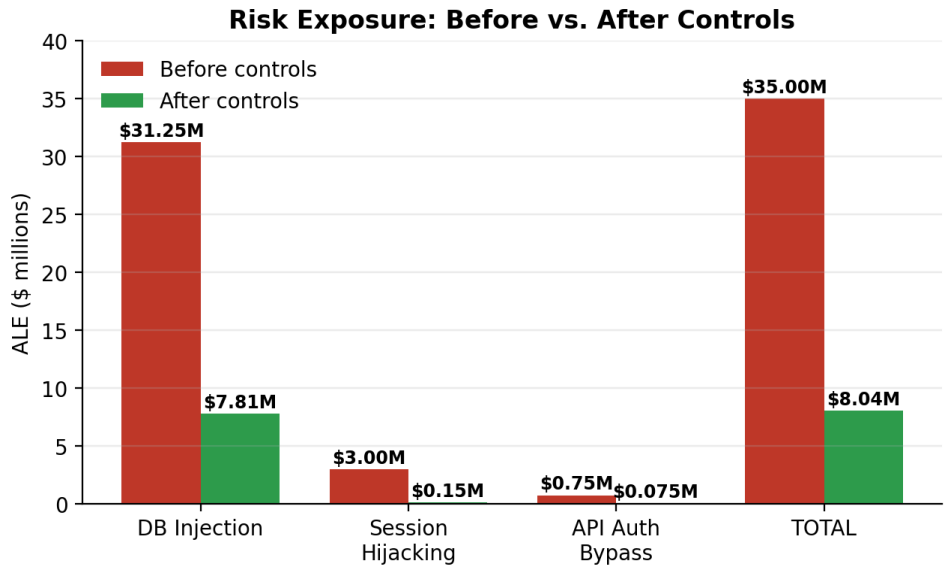


Figure 6. Risk exposure before vs. after controls

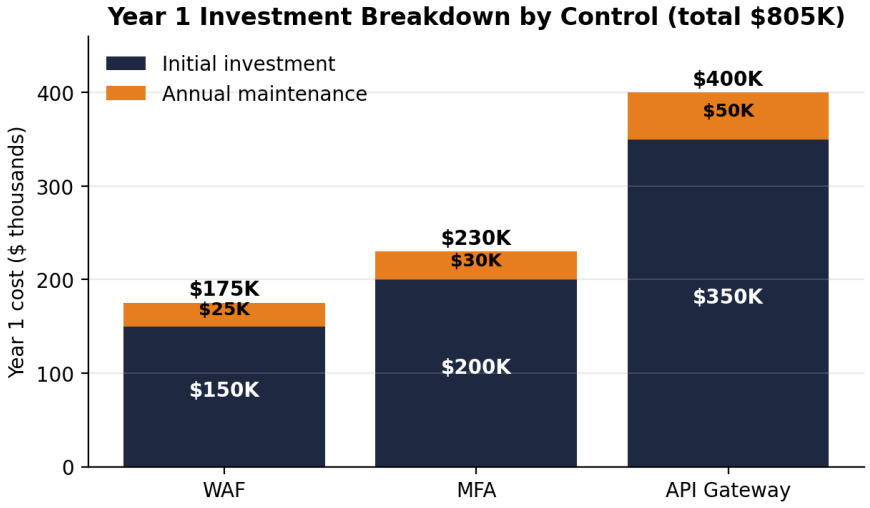


Figure 7. Year 1 investment breakdown by control (initial + maintenance)

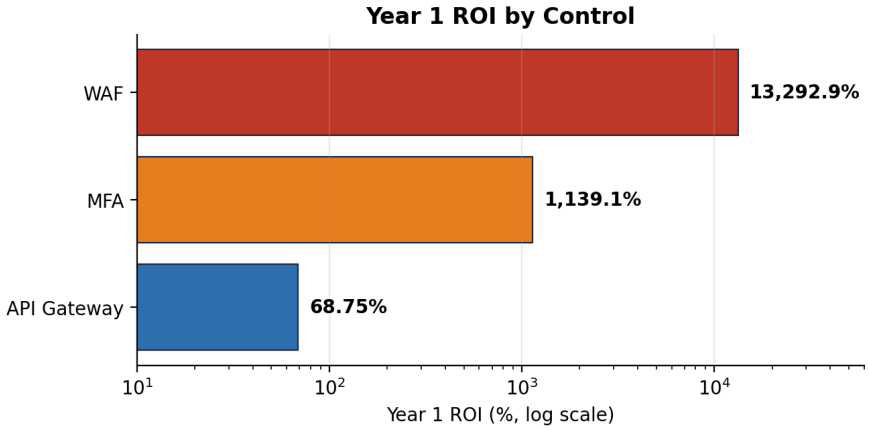


Figure 8. Year 1 ROI by control (log scale)

Task 7: Risk Treatment Timeline (Gantt Chart)

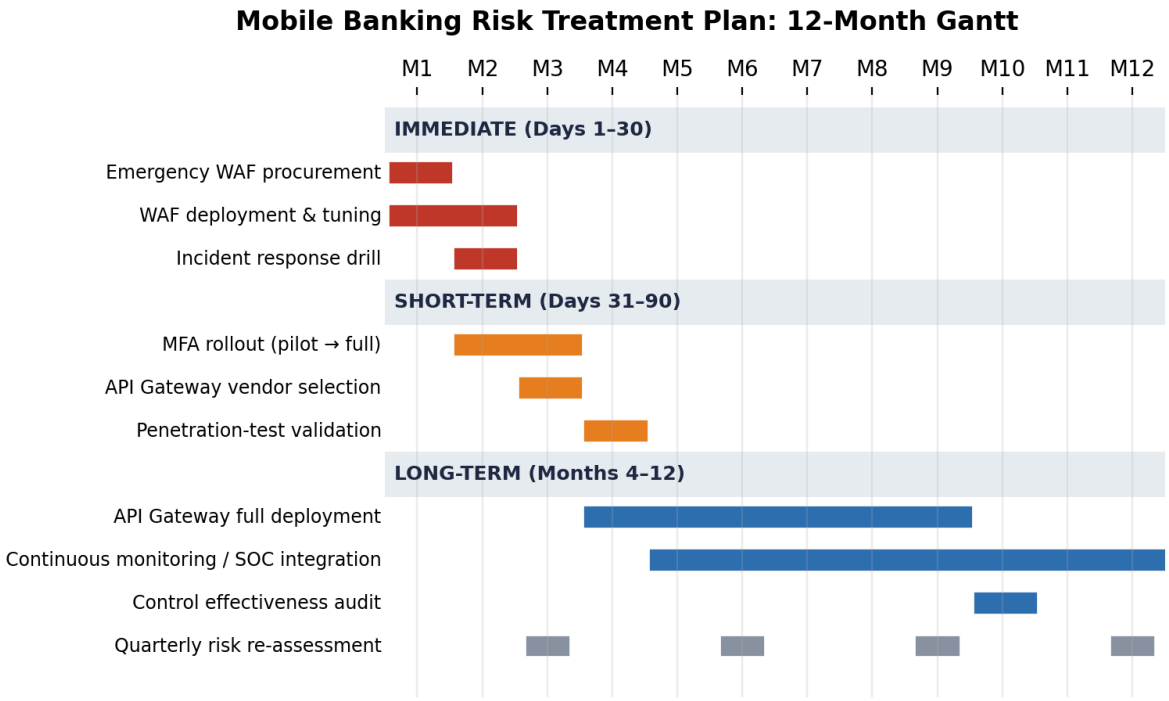


Figure 9. 12-month risk treatment plan

Phase	Window	Key actions	Why this order
Immediate	Days 1-30	Emergency WAF procurement; WAF deployment and tuning; incident response drill	WAF addresses 89% of total risk and pays back in about 2 days
Short-term	Days 31-90	MFA rollout (pilot to full); API Gateway vendor selection; penetration-test validation	MFA has high ROI and stops account takeover with moderate effort
Long-term	Months 4-12	API Gateway full deployment; SOC monitoring integration; quarterly risk reviews; control effectiveness audit	Largest upfront cost and longest integration, but lowest current ALE

Bonus Challenge: Combined Risk-Control Matrix

Current vs. residual risk

Vulnerability	Current ALE	Control	Effectiveness	Residual ALE	Risk reduced
Database Injection	$31,250,000	WAF	75%	$7,812,500	$23,437,500
Session Hijacking	$3,000,000	MFA	95%	$150,000	$2,850,000
API Auth Bypass	$750,000	API Gateway	90%	$75,000	$675,000
Total	**$35,000,000**	-	**77.0%**	**$8,037,500**	**$26,962,500**

Control cost vs. risk-reduction benefit

Control	Yr 1 cost	Risk reduction	Benefit : cost	Net benefit
WAF	$175,000	$23,437,500	133.9 : 1	$23,262,500
MFA	$230,000	$2,850,000	12.4 : 1	$2,620,000
API Gateway	$400,000	$675,000	1.69 : 1	$275,000
Combined	**$805,000**	**$26,962,500**	**33.5 : 1**	**$26,157,500**

Optimal selection under budget constraints

Budget scenario	Recommended controls	Total cost	Risk reduced	Residual ALE
Up to $200K	WAF only	$175,000	$23,437,500	$11,562,500
Up to $450K	WAF + MFA	$405,000	$26,287,500	$8,712,500
Up to $850K (full)	WAF + MFA + API Gateway	$805,000	$26,962,500	$8,037,500

> **Optimal recommendation:** Deploy **all three** controls. The combined $805K yields a 33.5 : 1 benefit-cost ratio and cuts residual risk to $8.04M. If the budget is capped at $450K, **WAF + MFA** captures 97.5% of the achievable risk reduction for about 50% of the cost.

Documentation Notes

1. **ARO = exploit probability** (annualized), per the lab framing.
2. **SLE for Database Injection** uses full record count × cost per record ($125M) as the single-incident exposure. This is a conservative, worst-case full breach.
3. **Control effectiveness is applied to the targeted ALE only**, with no double-counting: WAF → Database Injection; MFA → Session Hijacking; API Gateway → API Bypass.
4. **ROI is Year 1** (initial + first-year maintenance vs. first-year risk reduction). Multi-year ROI would be higher because risk reduction persists while only maintenance recurs.
5. **Payback period** = initial investment ÷ (annual risk reduction - annual maintenance).